EduOS Atlas Mock Examination

Pearson Edexcel International GCSE
Biology — Mark Scheme

Paper 1: Core Biology (4BI1/1C)

Mock Set 1

Total marks: 110 | Time allowed: 2 hours

Notes for Markers

- Award marks only when the marking point is clearly demonstrated in the candidate's response.
- Accept alternative scientific terminology that conveys the same meaning.
- Do not penalise minor spelling errors provided the intended word is unambiguous.
- Allow credit for correct working where a final answer is wrong due to arithmetic error.
- For extended response questions, use the indicative content as a guide, not a checklist.
- Quality of Written Communication (QWC) is assessed on questions marked with an asterisk (*).

SECTION A — Multiple Choice (10 marks)

Question	Marking Points / Acceptable Answers	Marks
1	B — It controls what enters and leaves the cell	**1**
2	C — Chloroplast	**1**
3	C — Circulatory and respiratory systems	**1**
4	C — Amylase	**1**
5	B — From high water potential to low water potential	**1**
6	C — Aorta	**1**
7	B — The enzyme has been denatured	**1**
8	B — Movement from high to low concentration along a concentration gradient	**1**
9	C — Pancreas	**1**
10	A — Oesophagus → Stomach → Small intestine → Large intestine	**1**

SECTION B — Short Answer and Structured Questions (100 marks)

Question 11: Cells and Microscopy

Question	Marking Points / Acceptable Answers	Marks
11(a)	Controls what substances enter and leave the cell (1) Accept: selectively permeable / acts as a barrier	**1**
11(b)	Any TWO of (1 mark each): • Plant cell has cell wall / animal cell does not (1) • Plant cell has chloroplasts / animal cell does not (1) • Plant cell has a large permanent vacuole / animal cell does not (1) Note: accept converse statements	**2**
11(c)	Magnification = image size ÷ actual size (1) = 30 ÷ 0.03 = 1000× (1) Allow 1 mark for correct formula used even if arithmetic error	**2**
11(d)	Level 3 (3 marks): Clear, coherent explanation of fluid mosaic model with named components and function Level 2 (2 marks): Some relevant detail e.g. phospholipid bilayer and channel proteins mentioned Level 1 (1 mark): General reference to membrane structure or partial permeability Indicative content: • Phospholipid bilayer — hydrophilic heads face outward, hydrophobic tails inward • Proteins embedded in the bilayer — some form channels/carriers • Small/non-polar molecules (e.g. O_2 , CO_2) can diffuse freely • Larger/polar molecules require protein channels or active transport • Selective permeability allows regulation of cell contents	**3**

Question 12: Enzymes

Question	Marking Points / Acceptable Answers	Marks

12(a)	pH 8 (1) Accept: between 7 and 9 if justified	**1**
12(b)	Rate = 1/time × 1000 = 1/560 × 1000 (1) = 1.79 (units: min^{-1} × 1000) (1) Accept 1.8 or values rounded to 1 d.p.	**2**
12(c)	Above the optimum pH, the rate decreases (1) Increasing H^+/OH^- ions alter ionic interactions within the enzyme (1) The active site changes shape (1) / enzyme becomes denatured The substrate can no longer fit the active site (1) Note: do not award mark for 'enzyme is killed' Award max 3	**3**
12(d)	Temperature affects enzyme activity (1) Higher temperature → higher rate of reaction / kinetic energy (1) Results at the same pH but different temperatures are not comparable (1) Temperature is a confounding variable, reducing reliability (1) Award max 2	**2**

Question 13: Human Nutrition and Digestion

Question	Marking Points / Acceptable Answers	Marks
13(a)	Enzyme: Salivary amylase (1) Food molecule: Starch (1) Accept: carbohydrate for starch	**2**
13(b)	Provides an acidic pH optimal for pepsin activity (1) Kills harmful bacteria in food (1) Award max 2	**2**
13(c)	Any THREE of (1 mark each): • Villi increase surface area (1) • Microvilli (brush border) on villi further increase SA (1) • Good blood supply / capillary network maintains concentration gradient (1) • Thin epithelium — short diffusion distance (1) • Lacteals for absorption of fatty acids/glycerol (1)	**3**
13(d)	Lipase breaks down fats into fatty acids and glycerol (1) Without lipase, fats cannot be digested / broken down (1) Fatty acids and glycerol cannot be absorbed (1) Accept: fats would pass through undigested and not provide energy (1) Award max 3	**3**

Question 14: Respiration

Question	Marking Points / Acceptable Answers	Marks
14(a)	As temperature decreases from 25°C to 5°C, oxygen consumption increases (1) Accept: inverse relationship / negative correlation (1) Reference to specific values, e.g. 8.7 at 25°C and 18.2 at 5°C (1) Award max 2	**2**

14(b)	Mammal is warm-blooded (endotherm) (1) At low temperatures, more energy/heat needed to maintain body temperature (1) Therefore metabolic rate and oxygen consumption are higher (1) Award max 2	**2**
14(c)	glucose → lactic acid + energy (1)	**1**
14(d)	Oxygen is used for energy production so supply cannot meet demand during intense exercise (1) Glucose is converted to lactic acid in muscles (1) Lactic acid accumulates causing muscle fatigue/cramp (1) After exercise, lactic acid is transported to liver (1) Oxygen debt is repaid — lactic acid is oxidised/converted to CO_2 and water (1) Award max 4	**4**

Question 15: Photosynthesis

Question	Marking Points / Acceptable Answers	Marks
15(a)	carbon dioxide + water → glucose + oxygen (1) [all four correct for mark]	**1**
15(b)(i)	As light intensity increases, rate of photosynthesis increases (1) Rate levels off at approximately 600–800 lux (1) Award 1 mark for description, 1 for plateau/levelling off	**2**
15(b)(ii)	Another factor becomes limiting (1) e.g. CO_2 concentration or temperature is insufficient (1) Amount of chlorophyll / enzyme is a limiting factor (1) Award max 2	**2**
15(c)	Level 3 (4 marks): Coherent explanation linking all three with light-dependent and light-independent stages Level 2 (2–3 marks): Two of the three components explained with some accuracy Level 1 (1 mark): One correct statement Indicative content: • Chlorophyll absorbs light energy for photosynthesis • Light splits water into H^+ and O_2 (photolysis) — O_2 released • Light energy used to produce ATP • CO_2 fixed in light-independent reactions (Calvin cycle) • CO_2 + H^+ → glucose (using ATP from light reactions)	**4**

Question 16: Transport in Plants

Question	Marking Points / Acceptable Answers	Marks
16(a)	X: Xylem (1) Y: Phloem (1)	**2**
16(b)	Water evaporates from spongy/palisade mesophyll cells (1) Water vapour diffuses out through stomata along a concentration gradient (1) This creates a water potential gradient pulling water up through xylem (1) Accept: transpiration pull / cohesion-tension mechanism (1) Award max 3	**3**
16(c)(i)	Moving air, low humidity (1)	**1**
16(c)(ii)	Moving air removes water vapour from around stomata (1) Maintains/increases the diffusion gradient for water vapour (1) Rate of evaporation from leaf increases (1) Award max 2	**2**

Question 17: Inheritance

Question	Marking Points / Acceptable Answers	Marks
17(a)	Parent 1: Tt Parent 2: Tt (1) [both must be correct]	**1**
17(b)	Correct Punnett square: TT, Tt, Tt, tt (1) All four offspring correctly placed (1) Award 1 for all four genotypes correct; 1 additional if all placed correctly	**2**
17(c)	3 tall : 1 dwarf (1) [accept 3:1 tall to dwarf]	**1**
17(d)	Perform a test cross — cross the tall plant with a dwarf plant (tt) (1) If all offspring are tall → plant is homozygous (TT) (1) If offspring are 50% tall, 50% dwarf → plant is heterozygous (Tt) (1) Award max 3	**3**
17(e)	Genotype: the genetic make-up of an organism / alleles it carries (1) Phenotype: the observable characteristics / physical expression of the genotype (1)	**2**

Question 18: Ecology and the Environment

Question	Marking Points / Acceptable Answers	Marks
18(a)	Oak tree (1)	**1**
18(b)	Blue tit population would decrease (1) Because caterpillars are the food source for blue tits (1) Less food → fewer blue tits survive / less reproduction (1) Award max 2	**2**
18(c)(i)	Efficiency = (energy at next level ÷ energy at previous level) × 100 (1) = (4200 ÷ 40 000) × 100 = 10.5% (1) Award 1 for correct formula; 1 for correct calculation	**2**
18(c)(ii)	Energy lost as heat during respiration (1) Energy used for movement/excretion/egestion (not available to next	**3**

	level) (1) Not all biomass is consumed or digested (1) Award max 3	

Question 19: Human Circulatory System

Question	Marking Points / Acceptable Answers	Marks
19(a)	Transport of substances (oxygen/nutrients/hormones/waste) around the body (1)	**1**
19(b)(i)	Pulmonary vein (1)	**1**
19(b)(ii)	Left ventricle pumps blood to the whole body / systemic circulation (1) Right ventricle pumps blood only to the lungs (shorter distance) (1) Greater muscle mass generates higher pressure needed for systemic circulation (1) Award max 2	**2**
19(c)	Arteries: thick muscular wall / elastic tissue / small lumen / carry blood away from heart (1) Veins: thin wall / large lumen / valves to prevent backflow / carry blood to heart (1) Capillaries: one cell thick wall / large SA:V / exchange of substances with tissues (1) Award max 4 — must compare, not just describe one	**4**
19(d)*	Level 3 (5–6 marks): Balanced evaluation of multiple modifiable and non-modifiable risk factors with clear, accurate scientific explanation Level 2 (3–4 marks): Describes 2–3 risk factors with some evaluation Level 1 (1–2 marks): Limited, one-sided or inaccurate content Indicative content — modifiable: • Smoking — nicotine raises blood pressure; CO reduces O_2 capacity • Diet high in saturated fats → atherosclerosis / plaque formation • Obesity → increased strain on heart • Physical inactivity • High alcohol intake Non-modifiable: • Age — risk increases with age • Genetics / family history • Sex — males at higher risk at younger ages • Ethnicity	**6**

Question 20: Practical Skills and Scientific Enquiry

Question	Marking Points / Acceptable Answers	Marks
20(a)	Rate of transpiration / distance bubble moves in potometer (per unit time) (1)	**1**
20(b)	Any TWO of (1 each): • Temperature • Humidity (if not the variable being tested) • Wind speed (if not being tested) • Size of leaf / number of leaves • Time of measurement	**2**
20(c)	Air bubble introduced into capillary tube (1)	**3**

	Distance bubble moves in a set time is measured (1) Greater distance = higher rate of transpiration (1) Process repeated at different light intensities (1) Award max 3	
20(d)(i)	300 lux mean = (2.1 + 2.3) ÷ 2 = 2.2 (1) 500 lux mean = (3.0 + 2.9) ÷ 2 = 2.95 (1) Award 1 mark for each correct mean	**1**
20(d)(ii)	Anomalous result: Trial 1 at 100 lux (1.2) — lower than expected / inconsistent with trend (1) Possible reason: air bubble may have moved back / leaf damaged / equipment not sealed (1) Award max 2	**2**
20(e)	Repeat each measurement more times (e.g. three trials) and calculate a mean (1) / Use the same plant throughout / Allow plant to acclimatise at each light intensity (1) Award 1 for any valid improvement	**1**

Total for Paper 1: 110 marks

End of Mark Scheme — Paper 1 | Created by EduOS Campus

EduOS Campus Mock Examination

Pearson Edexcel International GCSE
Biology — Mark Scheme

Paper 2: Biology Only (4BI1/2B)

Mock Set 1

Total marks: 70 | Time allowed: 1 hour 15 minutes

Notes for Markers

- Award marks only when the marking point is clearly demonstrated in the candidate's response.
- Accept alternative scientific terminology that conveys the same meaning.
- Do not penalise minor spelling errors provided the intended word is unambiguous.
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Question 1: Nervous System and Coordination

Question	Marking Points / Acceptable Answers	Marks
1(a)	Speeds up / insulates the nerve impulse (1) Accept: allows saltatory conduction (1) Do not accept: 'protects the axon' alone	**1**
1(b)	Nerve impulse arrives at pre-synaptic knob (1) Calcium ions enter, triggering vesicles to fuse with membrane (1) Neurotransmitter released into synaptic cleft by exocytosis (1) Neurotransmitter diffuses across cleft and binds to receptor on post-synaptic membrane (1) Generates new impulse in post-synaptic neurone (1) Neurotransmitter broken down/re-uptaken by pre-synaptic neurone (1) Award max 4	**4**
1(c)	Receptor (pain receptor in skin) (1) Sensory neurone carries impulse to spinal cord (1) Relay neurone in spinal cord (1) Motor neurone carries impulse to effector (muscle) (1) Muscle contracts, hand withdrawn (1) Award max 3	**3**
1(d)	They are fast / do not require conscious processing (1) Protect the organism from injury/damage before thinking is involved (1) Award max 2	**2**

Question 2: Hormones and Homeostasis

Question	Marking Points / Acceptable Answers	Marks
2(a)	Hormone 1: Insulin (1) Hormone 2: Glucagon (1) Both required for 2 marks	**2**
2(b)(i)	Approximately 45 minutes after the meal (accept 30–60 minutes) (1)	**1**
2(b)(ii)	Beta cells in pancreas detect high blood glucose (1) Insulin secreted into blood (1) Insulin binds to receptors on liver cells (1) Liver converts excess glucose to glycogen (glycogenesis) (1) Blood glucose concentration returns to normal set point (~5 mmol/L) (1) Award max 4	**4**
2(c)*	Level 3 (5–6 marks): Detailed, balanced comparison of both types with cause, symptoms and treatment clearly distinguished Level 2 (3–4 marks): Compares types with some accuracy; may lack balance or depth Level 1 (1–2 marks): Limited, mostly factual statements, little evaluation Indicative content: Type 1 — autoimmune destruction of beta cells; no insulin produced; requires insulin injections; juvenile onset; not preventable Type 2 — cells become resistant to insulin; lifestyle and genetic factors; managed by diet, exercise, oral medication or insulin; adult	**6**

	onset; preventable by lifestyle changes Similarities: both result in high blood glucose; both require monitoring; both can lead to complications (kidney damage, blindness, nerve damage) Differences: cause (autoimmune vs. lifestyle), treatment (always insulin vs. diet first), onset age	

Question 3: Reproduction and Fertilisation

Question	Marking Points / Acceptable Answers	Marks
3(a)	Any TWO of (1 mark each): • Tail/flagellum for motility / swimming (1) • Many mitochondria in midpiece to provide ATP for movement (1) • Acrosome contains enzymes to penetrate egg membrane (1) • Streamlined / small head to reduce resistance (1) • Contains haploid nucleus (1)	**2**
3(b)	Sperm reaches egg and contacts zona pellucida (1) Acrosome reaction — enzymes released to digest zona pellucida (1) Sperm head enters egg (1) Cortical reaction — zona pellucida hardens to prevent polyspermy (1) Nuclei of egg and sperm fuse (1) Diploid zygote formed (1) Award max 3	**3**
3(c)(i)	FSH stimulates development of a follicle in the ovary (1) Stimulates oestrogen production (1) Award max 2	**2**
3(c)(ii)	Corpus luteum degenerates if no fertilisation (1) Progesterone production falls (1) Uterus lining breaks down → menstruation begins (1) Award max 2	**2**

Question 4: Microorganisms, Disease and Immunity

Question	Marking Points / Acceptable Answers	Marks
4(a)	Erythromycin (antibiotic C) (1) Because it produces the largest inhibition zone (24 mm) (1)	**2**
4(b)	Staphylococcus aureus has developed resistance to Penicillin (1) (MRSA-type resistance) Accept: bacterium has an enzyme (penicillinase/beta-lactamase) that breaks down penicillin (1)	**1**
4(c)	Antibiotics target bacterial cell structures (cell wall/ribosomes) (1) Viruses do not have cell walls or their own ribosomes (1) Viruses replicate inside host cells using host machinery (1) Award max 2	**2**
4(d)*	Level 3 (5–6 marks): Detailed account of all three: phagocytes, lymphocytes, antibodies, with logical sequence and specific terminology Level 2 (3–4 marks): Describes two of the three with reasonable accuracy Level 1 (1–2 marks): Vague or incomplete description	**6**

	Indicative content: • Phagocytes engulf and digest bacteria by phagocytosis (non-specific) • Antigens on bacteria recognised as foreign • Lymphocytes produce specific antibodies complementary to antigens • Antibodies bind to antigens — agglutination, neutralisation, or opsonisation • B-lymphocytes produce plasma cells for antibody production • Memory cells formed — faster response on subsequent infection • T-lymphocytes destroy infected cells (cell-mediated immunity)	

Question 5: Evolution, Natural Selection and Classification

Question	Marking Points / Acceptable Answers	Marks
5(a)	Before the Industrial Revolution, tree bark was light-coloured; light moths were better camouflaged (1) Industrial pollution darkened tree bark with soot (1) Light-coloured moths became more visible to predators and were eaten more (1) Melanic moths had better camouflage on dark bark — more survived (1) Melanic moths reproduced and passed on the allele for dark colouration (1) Over generations, the frequency of the melanic allele increased in the population (1) Award max 4	**4**
5(b)	Prediction: frequency of melanic form decreased / light form increased (1) Reasoning: bark became lighter, so light moths better camouflaged again (1) Melanic moths more visible → eaten more → fewer survive to reproduce (1) Award max 3	**3**
5(c)	A species is a group of organisms that can interbreed to produce fertile offspring (1) They share common characteristics and a common ancestry (1) Award max 2	**2**
5(d)	Each organism is given a two-part name: genus and species (1) Advantage: allows scientists worldwide to refer to the same organism without confusion (1) / avoids problems with common names that vary by country/language (1) Award max 3 — one for definition, one for advantage with explanation	**3**

Total for Paper 2: 70 marks

End of Mark Scheme — Paper 2 | Created by EduOS Campus